03 IAM User Permission Lab - KirkYagami

IAM User Permissions Lab - S3 Access Control

Lab Overview

Objective: Create an IAM user with no permissions, test S3 access, then gradually add permissions to understand AWS security model.

Duration: 30-45 minutes

Prerequisites:

- AWS Account with admin access
- AWS CLI installed on local machine
- Basic command line knowledge

Lab Architecture

```
AWS Account (Root/Admin User)
    ↓
IAM User (test-user) - No Permissions
    ↓
Test S3 Access (Should Fail)
    ↓
Add S3 Read Permissions
    ↓
Test Again (Should Succeed)
```

Part 1: Create IAM User with No Permissions

Step 1: Access IAM Console

1. Log into AWS Console with admin credentials
2. Navigate to **IAM** service
3. Click on **Users** in the left sidebar
4. Click **Create user** button

Step 2: Configure User Details

1. **User name:** `test-user-lab`
2. **Provide user access to AWS Management Console:** ✅ Check this
3. **Console password:** Choose "Custom password"
   - Password: `TempPassword123!`
4. **Users must create a new password at next sign-in:** ✅ Check this
5. Click **Next**

Step 3: Set Permissions (Important!)

1. **Permission options:** Select "Add user to group"
2. **DO NOT** select any existing groups
3. **DO NOT** attach any policies directly
4. **DO NOT** copy permissions from existing user
5. Click **Next**

Step 4: Review and Create

1. Review the configuration:
   - User name: `test-user-lab`
   - No groups assigned
   - No policies attached
2. Click **Create user**

Step 5: Generate Access Keys

1. Click on the newly created user `test-user-lab`
2. Go to **Security credentials** tab
3. Scroll down to **Access keys** section
4. Click **Create access key**
5. Select **Command Line Interface (CLI)**
6. Check "I understand the above recommendation"
7. Click **Next**
8. **Optional**: Add description tag
9. Click **Create access key**
10. ⚠️ **IMPORTANT**: Copy and save:
    - Access Key ID
    - Secret Access Key

Part 2: Configure AWS CLI with New User

Step 1: Create New AWS CLI Profile

```
aws configure --profile test-user-lab
```

Step 2: Enter Credentials

When prompted, enter:

- **AWS Access Key ID:** `[Your copied Access Key ID]`
- **AWS Secret Access Key:** `[Your copied Secret Access Key]`
- **Default region name:** `us-east-1` (or your preferred region)
- **Default output format:** `json`

Step 3: Verify Profile Creation

```
aws configure list --profile test-user-lab
```

Expected output:

```
      Name                    Value             Type    Location
      ----                    -----             ----    --------
   profile                test-user-lab           manual    --profile
access_key     ****************XXXX shared-credentials-file
secret_key     ****************XXXX shared-credentials-file
    region                us-east-1      config-file    ~/.aws/config
```

Part 3: Test S3 Access (Should Fail)

Step 1: Test S3 Bucket Listing

```
aws s3 ls --profile test-user-lab
```

Expected Result: ❌ Access Denied Error

```
An error occurred (AccessDenied) when calling the ListBuckets operation: Access
Denied
```

Step 2: Try Creating a Bucket

```
aws s3 mb s3://test-bucket-lab-$(date +%s) --profile test-user-lab
```

Expected Result: ❌ Access Denied Error

Step 3: Document the Results

- **S3 List Buckets:** ❌ Failed
- **S3 Create Bucket:** ❌ Failed
- **Reason:** User has no permissions attached

Part 4: Add S3 Read-Only Permissions

Step 1: Return to IAM Console

1. Go to **IAM** → **Users** → **test-user-lab**
2. Click on **Permissions** tab
3. Click **Add permissions** button
4. Select **Attach policies directly**

Step 2: Search and Attach S3 Read Policy

1. In the search box, type: `AmazonS3ReadOnlyAccess`
2. Check the box next to **AmazonS3ReadOnlyAccess**
3. Review the policy permissions (expand to see details)
4. Click **Next**
5. Click **Add permissions**

Step 3: Verify Policy Attachment

- User should now show **AmazonS3ReadOnlyAccess** in permissions tab
- Policy allows `s3:Get*` and `s3:List*` actions

Part 5: Test S3 Access Again (Should Succeed)

Step 1: Test S3 Bucket Listing

```
aws s3 ls --profile test-user-lab
```

Expected Result: ✅ Success - List of S3 buckets

```
2024-08-06 10:30:45 my-bucket-1
2024-08-06 11:15:22 my-bucket-2
```

Step 2: Test Reading Bucket Contents

```
aws s3 ls s3://[your-existing-bucket-name] --profile test-user-lab
```

Expected Result: ✅ Success - List of objects in bucket

Step 3: Test Write Operations (Should Still Fail)

```
aws s3 mb s3://test-write-bucket-$(date +%s) --profile test-user-lab
```

Expected Result: ❌ Still fails - No write permissions

Part 6: Upgrade to Full S3 Access

Step 1: Remove Read-Only Policy

1. Go to **IAM** → **Users** → **test-user-lab** → **Permissions**
2. Find **AmazonS3ReadOnlyAccess** policy
3. Click the **X** next to it to remove
4. Confirm removal

Step 2: Add Full S3 Access Policy

1. Click **Add permissions**

2. Select **Attach policies directly**
3. Search for: `AmazonS3FullAccess`
4. Check **AmazonS3FullAccess** policy
5. Click **Next** → **Add permissions**

Step 3: Test Full Access

```
# Create bucket (should now work)
aws s3 mb s3://test-full-access-$(date +%s) --profile test-user-lab

# Upload a file
echo "Hello World" > test-file.txt
aws s3 cp test-file.txt s3://[your-bucket-name]/ --profile test-user-lab

# Delete the test file
aws s3 rm s3://[your-bucket-name]/test-file.txt --profile test-user-lab
```

Expected Result: ✅ All operations should succeed

Lab Results Summary

Permission Level	S3 List	S3 Read	S3 Write	S3 Delete
No Permissions	❌	❌	❌	❌
S3 Read-Only	✅	✅	❌	❌
S3 Full Access	✅	✅	✅	✅

Key Learning Points

IAM Security Model

- **Principle of Least Privilege:** Start with no permissions, add only what's needed
- **Explicit Deny:** AWS denies all actions unless explicitly allowed
- **Policy Inheritance:** Users inherit permissions from groups and directly attached policies

S3 Permission Types

- **List Permissions:** `s3:ListBucket`, `s3:ListAllMyBuckets`
- **Read Permissions:** `s3:GetObject`, `s3:GetObjectVersion`
- **Write Permissions:** `s3:PutObject`, `s3:PutObjectAcl`
- **Delete Permissions:** `s3:DeleteObject`, `s3:DeleteBucket`

Best Practices Observed

1. **Never use root credentials** for daily operations
2. **Create specific users** for different applications/purposes
3. **Use groups** for managing permissions at scale
4. **Regularly audit** user permissions
5. **Rotate access keys** periodically

Cleanup Instructions

Step 1: Delete Test Resources

```
# Remove test file if created
rm test-file.txt

# Delete any test buckets created
aws s3 rb s3://[test-bucket-name] --force --profile test-user-lab
```

Step 2: Remove IAM User

1. Go to **IAM** → **Users** → **test-user-lab**
2. **Security credentials** tab → Delete access keys
3. **Permissions** tab → Remove all attached policies
4. Click **Delete** user
5. Type user name to confirm deletion

Step 3: Remove CLI Profile

```
aws configure list-profiles
# Remove the profile from ~/.aws/credentials and ~/.aws/config files
```

Troubleshooting

Common Issues

CLI Profile Not Working

- Check `aws configure list --profile test-user-lab`
- Verify access keys are correctly entered
- Ensure no extra spaces in keys

Still Getting Access Denied After Adding Permissions

- Wait 1-2 minutes for policy changes to propagate
- Check policy attachment in IAM console
- Verify using correct profile: `--profile test-user-lab`

Cannot Delete User

- Remove all access keys first
- Detach all policies and groups
- User must have no resources attached

Extension Exercises

1. **Create a Custom Policy** that allows only listing buckets but not reading objects
2. **Set up MFA** requirement for the IAM user
3. **Create an IAM Group** and add the user to it instead of direct policy attachment

4. **Test Cross-Region Access** by creating resources in different regions
5. **Implement Resource-Based Policies** using bucket policies

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Tags

#aws #iam #s3 #security #permissions #lab #hands-on